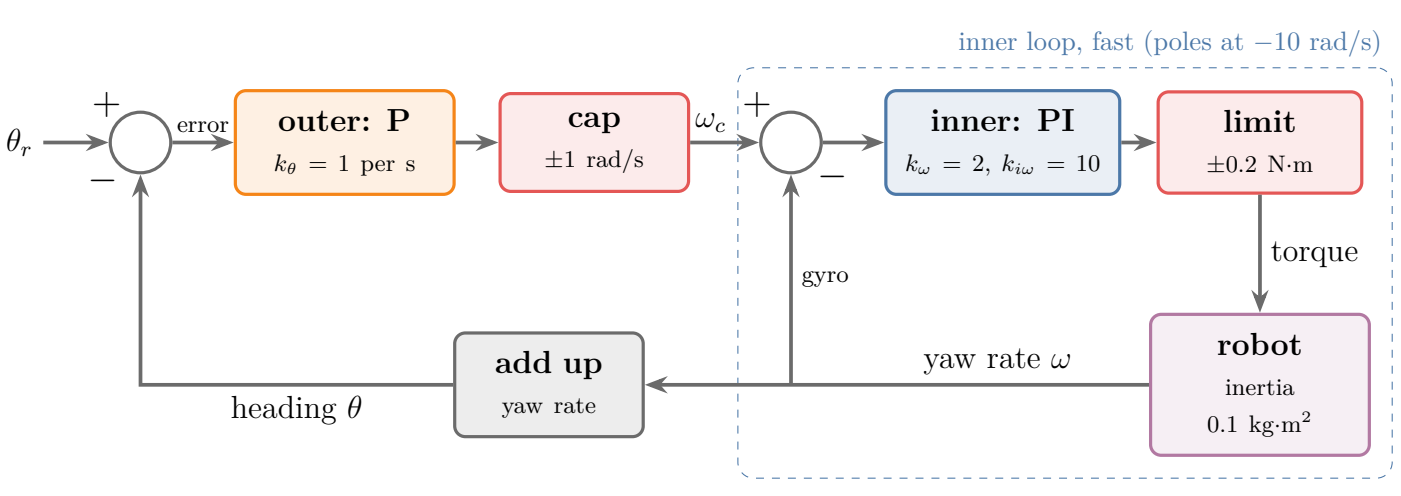

inner loop, fast (poles at −10 rad/s)
θ_r
+
−
error
outer: P
k_θ = 1 per s
cap
±1 rad/s
ω_c
+
−
inner: PI
k_ω = 2, $k_{i\omega}$ = 10
limit
±0.2 N·m
torque
gyro
robot
inertia
0.1 kg·m^2
yaw rate ω
add up
yaw rate
heading θ